

100 Days of Azure DevOps

Day 11 Handout — Working tree, index, commit DAG, remote — four areas, one SHA

2 - Azure Repos & Git Mastery · Personal lab only · Educational content · Not a sales pitch

Architecture A — four Git areas (the system map)

Area	On disk	Command that moves data
Working tree	Your files (notes/day11.md)	edit; git status shows unstaged
Index / staging	.git/index	git add notes/day11.md
Local repo	.git/objects commit/tree/blob	git commit -m "docs: ..."
Remote	Azure Repos refs/heads/*	git push -u origin feature/day11-notes
HEAD	Pointer to branch or detached SHA	git switch / git checkout
Ref	.git/refs/heads/main (41-byte SHA + newline)	git branch -v
Stash (optional)	.git/refs/stash	Do not stash as a backup strategy

Architecture B — object model (decide what a commit actually is)

Object	Contains	Why CI cares
Blob	File bytes, SHA-1 (or SHA-256 if enabled)	Content-addressed; rename is cheap
Tree	Names + modes + child SHAs	Pipeline checks out this tree
Commit	tree + parent(s) + author + message	Build reason: IndividualCI on that SHA
Tag	Annotated tag object or lightweight ref	Release tags later; not required today
Branch	Movable ref, not a copy of files	feature/* isolation without new repos
Merge vs fast-forward	Extra commit with 2 parents vs pointer move	Day 15; today prefer linear local merge

Architecture C — pitfalls that rewrite the wrong universe

Pitfall	What you see	Fix
Commit -a by reflex	Unrelated files in the SHA	git add -p / add the path; git status first
Push to work remote	origin URL is employer	git remote -v; origin must be personal Azure Repos
Empty commits as save points	Noise in git log --oneline	Commit when the note is a complete thought
Main for experiments	Broken README on default branch	git switch -c feature/day11-notes
Line-ending war	Whole-file diffs on Windows	core.autocrlf true on Win; .gitattributes later
git add . at repo root	os_user credentials in .env	No .env; Day 13 .gitignore

One-liner to remember

Git is time travel with extra anxiety — commits are save points; branches are parallel universes you can still mess up.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal clone of azure-100-labs only. origin = https://dev.azure.com/<org>/azure-100-labs/_git/azure-100-labs

- 1 git clone (if needed) && git remote -v. Confirm origin is the personal Azure Repos URL, not a work GitHub.
- 2 git status && git switch -c feature/day11-notes
- 3 mkdir notes (Windows: mkdir notes) && write notes/day11.md with the four-area map in your words.
- 4 git add notes/day11.md && git status (should show staged, not a pile of junk).
- 5 git commit -m "docs: day 11 git fundamentals notes"
- 6 git log --oneline --decorate --graph -5 && git push -u origin feature/day11-notes
- 7 Azure Repos → Branches → see feature/day11-notes. Optional: merge locally to main or wait for Day 14 PR.

Branch, commit one file, push with upstream tracking

```
git status
git switch -c feature/day11-notes
mkdir notes 2>nul
echo Day 11: working tree - index - commit - remote > notes/day11.md
git add notes/day11.md
git commit -m "docs: day 11 git fundamentals notes"
git log --oneline --decorate --graph -5
git push -u origin feature/day11-notes
```

Done checklist

- Can name working tree, index, local repo, remote
- feature/day11-notes exists on Azure Repos with one intentional commit
- git log --oneline shows the SHA
- origin is the personal azure-100-labs repo

Tomorrow — Day 12

Branching strategies